15. The electric field at a point $P(x, y, z)$ due to a point charge q located at the origin is given by the inverse square field

$$E = q\frac{\mathbf{r}}{\|\mathbf{r}\|^3},$$

where $r = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.

(a) Suppose S is a closed surface, S_a is a sphere $x^2 + y^2 + z^2 = a^2$ lying completely within S, and D is the region bounded between S and S_a. See **FIGURE 9.16.7**. Show that the outward flux of $\mathbf{E}$ for the region D is zero.

(b) Use the result of part (a) to prove **Gauss' law**:

$$\iint_S (\mathbf{E} \cdot \mathbf{n})\, dS = 4\pi q;$$

that is, the outward flux of the electric field $\mathbf{E}$ through *any* closed surface (for which the divergence theorem applies) containing the origin is $4\pi q$.

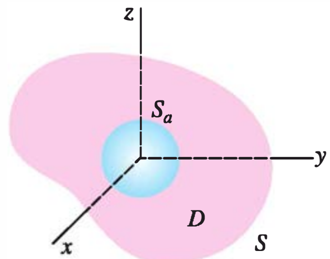


FIGURE 9.16.7 Region D for Problem 15(a)

16. Suppose there is a continuous distribution of charge throughout a closed and bounded region D enclosed by a surface S. Then the natural extension of Gauss' law is given by

$$\iint_S (\mathbf{E} \cdot \mathbf{n})\, dS = \iiint_D 4\pi\rho\, dV,$$

where $\rho(x, y, z)$ is the charge density or charge per unit volume.

(a) Proceed as in the derivation of the continuity equation (16) to show that div $\mathbf{E} = 4\pi\rho$.

(b) Given that $\mathbf{E}$ is an irrotational vector field, show that the potential ϕ for $\mathbf{E}$ satisfies Poisson's equation $\nabla^2\phi = 4\pi\rho$.

In Problems 17–21, assume that S forms the boundary of a closed and bounded region D.

17. If $\mathbf{a}$ is a constant vector, show that $\iint_S (\mathbf{a} \cdot \mathbf{n})\, dS = 0$.

18. If $\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$ and P, Q, and R have continuous second partial derivatives, prove that

$$\iint_S (\text{curl}\, \mathbf{F} \cdot \mathbf{n})\, dS = 0.$$

In Problems 19 and 20, assume that f and g are scalar functions with continuous second partial derivatives. Use the divergence theorem to establish **Green's identities**.

19. $$\iint_S (f\nabla g) \cdot \mathbf{n}\, dS = \iiint_D (f\nabla^2 g + \nabla f \cdot \nabla g)\, dV$$

20. $$\iint_S (f\nabla g - g\nabla f) \cdot \mathbf{n}\, dS = \iiint_D (f\nabla^2 g - g\nabla^2 f)\, dV$$

21. If f is a scalar function with continuous first partial derivatives, prove that

$$\iint_S f\mathbf{n}\, dS = \iiint_D \nabla f\, dV.$$

[*Hint*: Use (2) on $f\mathbf{a}$, where $\mathbf{a}$ is a constant vector, and Problem 27 in Exercises 9.7.]

22. The buoyancy force on a floating object is $\mathbf{B} = -\iint_S p\,\mathbf{n}\, dS$, where p is the fluid pressure. The pressure p is related to the density of the fluid $\rho(x, y, z)$ by a law of hydrostatics: $\nabla p = \rho(x, y, z)\mathbf{g}$, where $\mathbf{g}$ is the constant acceleration of gravity. If the weight of the object is $\mathbf{W} = m\mathbf{g}$, use the result of Problem 21 to prove Archimedes' principle, $\mathbf{B} + \mathbf{W} = \mathbf{0}$. See **FIGURE 9.16.8**.

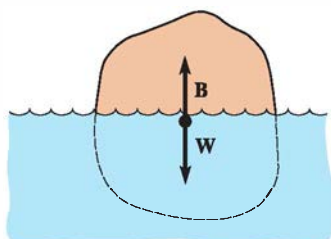


FIGURE 9.16.8 Floating object in Problem 22

9.17 Change of Variables in Multiple Integrals

Introduction In many instances it is either a matter of convenience or of necessity to make a substitution, or change of variable, in a definite integral $\int_a^b f(x)\, dx$ in order to evaluate it. If f is continuous on $[a, b]$, $x = g(u)$ has a continuous derivative, and $dx = g'(u)\, du$, then

$$\int_a^b f(x)\, dx = \int_c^d f(g(u))\, g'(u)\, du, \tag{1}$$

If the function g is one-to-one, then it has an inverse and so $c = g^{-1}(a)$ and $d = g^{-1}(b)$.

where the u-limits of integration c and d are defined by $a = g(c)$ and $b = g(d)$. There are three things that bear emphasizing in (1). To change the variable in a definite integral we replace x where it appears in the integrand by $g(u)$, we change the interval of integration $[a, b]$ on the x-axis to the corresponding interval $[c, d]$ on the u-axis, and we replace dx by a function multiple (namely, the derivative of g) of du. If we write $J(u) = dx/du$, then (1) has the form

$$\int_a^b f(x)\,dx = \int_c^d f(g(u))J(u)\,du. \tag{2}$$

For example, using $x = 2\sin\theta$, $-\pi/2 \le \theta \le \pi/2$, we get

$$\overset{x\text{-limits}}{\downarrow}\int_0^2 \overbrace{\sqrt{4 - x^2}}^{f(x)}\,dx = \overset{\theta\text{-limits}}{\downarrow}\int_0^{\pi/2} \overbrace{2\cos\theta}^{f(2\sin\theta)}\,\overbrace{(2\cos\theta)}^{J(\theta)}\,d\theta = 4\int_0^{\pi/2} \cos^2\theta\,d\theta = \pi.$$

Double Integrals Although changing variables in a multiple integral is not as straightforward as the procedure in (1), the basic idea illustrated in (2) carries over. To change variables in a double integral we need two equations such as

$$x = f(u, v), \qquad y = g(u, v). \tag{3}$$

To be analogous with (2), we expect that a change of variables in a double integral would take the form

$$\iint_R F(x, y)\,dA = \iint_S F(f(u, v), g(u, v))J(u, v)\,dA', \tag{4}$$

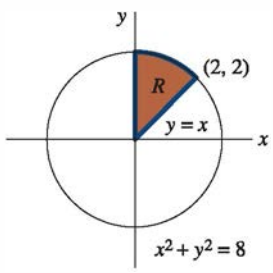
where S is the region in the uv-plane corresponding to the region R in the xy-plane and $J(u, v)$ is some function that depends on the partial derivatives of the equations in (3). The symbol dA' on the right side of (4) represents either $du\,dv$ or $dv\,du$.

In Section 9.11 we briefly discussed how to change a double integral $\iint_R F(x, y)\,dA$ from rectangular coordinates to polar coordinates. Recall that in Example 2 of that section the substitutions

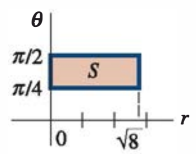
$$x = r\cos\theta, \qquad y = r\sin\theta \tag{5}$$

led to

$$\int_0^2\int_x^{\sqrt{8-x^2}} \frac{1}{5 + x^2 + y^2}\,dy\,dx = \int_{\pi/4}^{\pi/2}\int_0^{\sqrt{8}} \frac{1}{5 + r^2}\,r\,dr\,d\theta. \tag{6}$$



(a) Region R in xy-plane



(b) Region S in $r\theta$-plane

FIGURE 9.17.1 Region S is used to evaluate (6)

As we see in **FIGURE 9.17.1**, the introduction of polar coordinates changes the original region of integration R in the xy-plane to the more convenient rectangular region of integration S in the $r\theta$-plane. We note, too, that by comparing (4) with (6), we can identify $J(r, \theta) = r$ and $dA' = dr\,d\theta$.

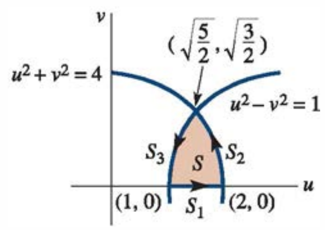
The change-of-variable equations in (3) define a **transformation** or **mapping** T from the uv-plane to the xy-plane. A point (x_0, y_0) in the xy-plane determined from $x_0 = f(u_0, v_0)$, $y_0 = g(u_0, v_0)$ is said to be an **image** of (u_0, v_0).

EXAMPLE 1 Image of a Region

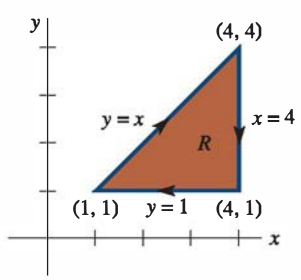
Find the image of the region S shown in **FIGURE 9.17.2(a)** under the transformation $x = u^2 + v^2$, $y = u^2 - v^2$.

SOLUTION We begin by finding the images of the sides of S that we have indicated by S_1, S_2, and S_3.

S_1: On this side $v = 0$ so that $x = u^2$, $y = u^2$. Eliminating u then gives $y = x$. Now imagine moving along the boundary from (1, 0) to (2, 0) (that is, $1 \le u \le 2$). The equations $x = u^2$, $y = u^2$ then indicate that x ranges from $x = 1$ to $x = 4$ and y ranges



(a)



(b)

FIGURE 9.17.2 Region R is the image of region S in Example 1 simultaneously from $y = 1$ to $y = 4$. In other words, in the xy-plane the image of S_1 is the line segment $y = x$ from $(1, 1)$ to $(4, 4)$.

S_2: On this boundary $u^2 + v^2 = 4$ and so $x = 4$. Now as we move from the point $(2, 0)$ to $(\sqrt{\frac{5}{2}}, \sqrt{\frac{3}{2}})$, the remaining equation $y = u^2 - v^2$ indicates that y ranges from $y = 2^2 - 0^2 = 4$ to $y = (\sqrt{\frac{5}{2}})^2 - (\sqrt{\frac{3}{2}})^2 = 1$. In this case the image of S_2 is the vertical line segment $x = 4$ starting at $(4, 4)$ and going down to $(4, 1)$.

S_3: Since $u^2 - v^2 = 1$, we get $y = 1$. But as we move on this boundary from $(\sqrt{\frac{5}{2}}, \sqrt{\frac{3}{2}})$, to $(1, 0)$, the equation $x = u^2 + v^2$ indicates that x ranges from $x = 4$ to $x = 1$. The image of S_3 is the horizontal line segment $y = 1$ starting at $(4, 1)$ and ending at $(1, 1)$.

The image of S is the region R given in Figure 9.17.2(b). ≡

Observe in Example 1 that as we traverse the boundary of S in the counterclockwise direction, the boundary of R is traversed in a clockwise manner. We say that the transformation of the boundary of S has *induced* an orientation on the boundary of R.

While a proof of the formula for changing variables in a multiple integral is beyond the level of this text, we will give *some* of the underlying assumptions that are made about the equations (3) and the regions R and S. We assume that

- The functions f and g have continuous first partial derivatives on S.
- The transformation is one-to-one.
- Each of the regions R and S consists of a piecewise-smooth simple closed curve and its interior.
- The determinant

$$\begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u} \tag{7}$$

  is not zero on S.

A transformation T is said to be **one-to-one** if each point (x_0, y_0) in R is the image under T of a unique point (u_0, v_0) in S. Put another way, no two points in S have the same image in R. With the restrictions that $r \geq 0$ and $0 \leq \theta \leq 2\pi$, the equations in (5) define a one-to-one transformation from the $r\theta$-plane to the xy-plane. The determinant in (7) is called the **Jacobian determinant**, or simply **Jacobian**, of the transformation T and is the key to changing variables in a multiple integral. The Jacobian of the transformation defined by the equations in (3) is denoted by the symbol

$$\frac{\partial(x, y)}{\partial(u, v)}.$$

Similar to the notion of a one-to-one function, a one-to-one transformation T has an **inverse transformation** T^{-1} such that (u_0, v_0) is the image under T^{-1} of (x_0, y_0). See **FIGURE 9.17.3**. If it is possible to solve (3) for u and v in terms of x and y, then the inverse transformation is defined by a pair of equations

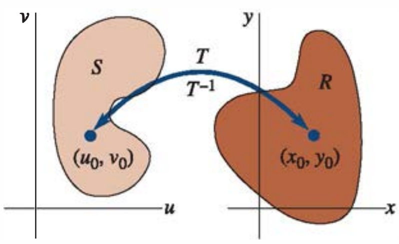


FIGURE 9.17.3 Transformation T and its inverse

$$u = h(x, y), \qquad v = k(x, y). \tag{8}$$

The Jacobian of the inverse transformation T^{-1} is

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \dfrac{\partial u}{\partial x} & \dfrac{\partial u}{\partial y} \\ \dfrac{\partial v}{\partial x} & \dfrac{\partial v}{\partial y} \end{vmatrix} \tag{9}$$

and is related to the Jacobian of the transformation T by

$$\frac{\partial(x, y)}{\partial(u, v)} \frac{\partial(u, v)}{\partial(x, y)} = 1. \tag{10}$$

EXAMPLE 2 Jacobian

The Jacobian of the transformation $x = r\cos\theta$, $y = r\sin\theta$ is

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \dfrac{\partial x}{\partial r} & \dfrac{\partial x}{\partial \theta} \\ \dfrac{\partial y}{\partial r} & \dfrac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r(\cos^2\theta + \sin^2\theta) = r. \quad \equiv$$

We now turn our attention to the main point of this discussion: how to change variables in a multiple integral. The idea expressed in (4) is valid; the function $J(u, v)$ turns out to be $|\partial(x, y)/\partial(u, v)|$. Under the assumptions made above, we have the following result:

Theorem 9.17.1 Change of Variables in a Double Integral

If F is continuous on R, then

$$\iint_R F(x, y)\,dA = \iint_S F(f(u, v), g(u, v)) \left|\frac{\partial(x, y)}{\partial(u, v)}\right| dA'. \tag{11}$$

Formula (3) of Section 9.11 for changing a double integral to polar coordinates is just a special case of (11) with

$$\left|\frac{\partial(x, y)}{\partial(r, \theta)}\right| = |r| = r,$$

since $r \geq 0$. In (6) then we have $J(r, \theta) = |\partial(x, y)/\partial(r, \theta)| = r$.

A change of variables in a multiple integral can be used for either a simplification of the integrand or a simplification of the region of integration. The actual change of variables used is often inspired by the structure of the integrand $F(x, y)$ or by equations that define the region R. As a consequence, the transformation is then defined by equations of the form given in (8); that is, we are dealing with the inverse transformation. The next two examples will illustrate these ideas.

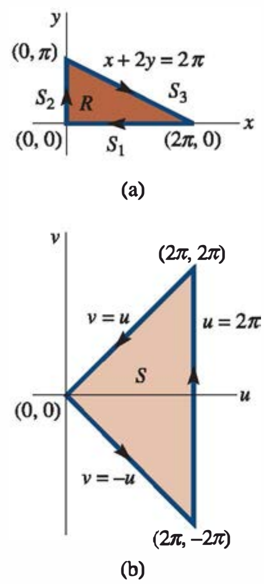


FIGURE 9.17.4 Region S is the image of region R in Example 3

EXAMPLE 3 Changing Variables in a Double Integral

Evaluate $\iint_R \sin(x + 2y)\cos(x - 2y)\,dA$ over the region R shown in **FIGURE 9.17.4(a)**.

SOLUTION The difficulty in evaluating this double integral is clearly the integrand. The presence of the terms $x + 2y$ and $x - 2y$ prompts us to define the change of variables $u = x + 2y$, $v = x - 2y$. These equations will map R onto a region S in the uv-plane. As in Example 1, we transform the sides of the region.

S_1: $y = 0$ implies $u = x$ and $v = x$ or $v = u$. As we move from $(2\pi, 0)$ to $(0, 0)$, we see that the corresponding image points in the uv-plane lie on the line segment $v = u$ from $(2\pi, 2\pi)$ to $(0, 0)$.

S_2: $x = 0$ implies $u = 2y$ and $v = -2y$, or $v = -u$. As we move from $(0, 0)$ to $(0, \pi)$, the corresponding image points in the uv-plane lie on the line segment $v = -u$ from $(0, 0)$ to $(2\pi, -2\pi)$.

S_3: $x + 2y = 2\pi$ implies $u = 2\pi$. As we move from $(0, \pi)$ to $(2\pi, 0)$, the equation $v = x - 2y$ shows that v ranges from $v = -2\pi$ to $v = 2\pi$. Thus, the image of S_3 is the vertical line segment $u = 2\pi$ starting at $(-2\pi, -2\pi)$ and going up to $(2\pi, 2\pi)$. See Figure 9.17.4(b).

Now, solving for x and y in terms of u and v gives

$$x = \frac{1}{2}(u + v), \quad y = \frac{1}{4}(u - v).$$

Therefore,
$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \dfrac{1}{2} & \dfrac{1}{2} \\ \dfrac{1}{4} & -\dfrac{1}{4} \end{vmatrix} = -\frac{1}{4}.$$

Hence, from (11) we find that

$$\iint_R \sin(x + 2y)\cos(x - 2y)\, dA = \iint_S \sin u \cos v \left|-\frac{1}{4}\right| dA'$$

$$= \frac{1}{4}\int_0^{2\pi}\int_{-u}^{u} \sin u \cos v \, dv\, du$$

$$= \frac{1}{4}\int_0^{2\pi} \sin u \sin v \Big]_{-u}^{u} du$$

$$= \frac{1}{2}\int_0^{2\pi} \sin^2 u\, du$$

$$= \frac{1}{4}\int_0^{2\pi} (1 - \cos 2u)\, du$$

$$= \frac{1}{4}\left[u - \frac{1}{2}\sin 2u\right]_0^{2\pi} = \frac{\pi}{2}.$$ ≡

EXAMPLE 4 Changing Variables in a Double Integral

Evaluate $\iint_R xy\, dA$ over the region R shown in **FIGURE 9.17.5(a)**.

SOLUTION In this case the integrand is fairly simple, but integration over the region R would be tedious since we would have to express $\iint_R xy\, dA$ as the sum of three integrals. (Verify this.)

The equations of the boundaries of R suggest the change of variables

$$u = \frac{y}{x^2}, \qquad v = xy. \tag{12}$$

Obtaining the image of R is a straightforward matter in this case, since the images of the curves that make up the four boundaries are simply $u = 1, u = 4, v = 1$, and $v = 5$. In other words, the image of the region R is the rectangular region S: $1 \le u \le 4, 1 \le v \le 5$. See Figure 9.17.5(b).

Now instead of trying to solve the equations in (12) for x and y in terms of u and v, we can compute the Jacobian $\partial(x, y)/\partial(u, v)$ by computing $\partial(u, v)/\partial(x, y)$ and using (10). We have

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \dfrac{\partial u}{\partial x} & \dfrac{\partial u}{\partial y} \\ \dfrac{\partial v}{\partial x} & \dfrac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} -\dfrac{2y}{x^3} & \dfrac{1}{x^2} \\ y & x \end{vmatrix} = -\frac{3y}{x^2},$$

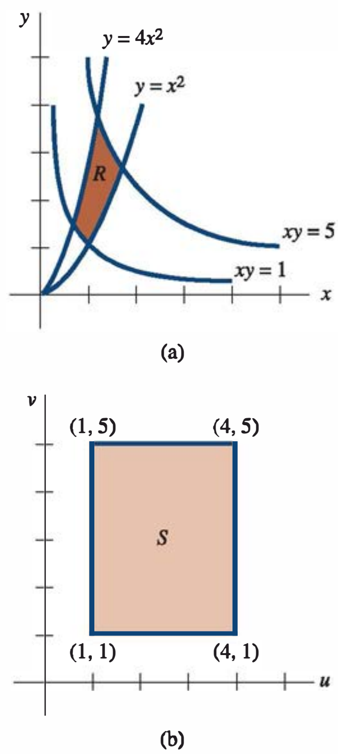


FIGURE 9.17.5 Region S is image of region R in Example 4

and so from (10),

$$\frac{\partial(x, y)}{\partial(u, v)} = \frac{1}{\dfrac{\partial(u, v)}{\partial(x, y)}} = -\frac{x^2}{3y} = -\frac{1}{3u}.$$

Hence,

$$\iint_R xy\, dA = \iint_S v\left|-\frac{1}{3u}\right| dA'$$

$$= \frac{1}{3}\int_1^4 \int_1^5 \frac{v}{u}\, dv\, du$$

$$= \frac{1}{3}\int_1^4 \frac{v^2}{2u}\bigg]_1^5 du$$

$$= 4\int_1^4 \frac{1}{u}\, du = 4 \ln u \bigg]_1^4 = 4 \ln 4. \quad \equiv$$

Triple Integrals To change variables in a triple integral, let

$$x = f(u, v, w), \qquad y = g(u, v, w), \qquad z = h(u, v, w),$$

be a one-to-one transformation T from a region E in uvw-space to a region D in xyz-space. If F is continuous on D, then

$$\iiint_D F(x, y, z)\, dV = \iiint_E F(f(u, v, w), g(u, v, w), h(u, v, w)) \left|\frac{\partial(x, y, z)}{\partial(u, v, w)}\right| dV'$$

where

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} & \dfrac{\partial x}{\partial w} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} & \dfrac{\partial y}{\partial w} \\ \dfrac{\partial z}{\partial u} & \dfrac{\partial z}{\partial v} & \dfrac{\partial z}{\partial w} \end{vmatrix}.$$

We leave it as an exercise for the reader to show that if T is the transformation from spherical to rectangular coordinates defined by

$$x = \rho \sin\phi \cos\theta, \qquad y = \rho \sin\phi \sin\theta, \qquad z = \rho\cos\phi, \tag{13}$$

then

$$\frac{\partial(x, y, z)}{\partial(\rho, \phi, \theta)} = \rho^2 \sin\phi.$$

9.17 Exercises

Answers to selected odd-numbered problems begin on page ANS-24.

1. Consider a transformation T defined by $x = 4u - v$, $y = 5u + 4v$. Find the images of the points (0, 0), (0, 2), (4, 0), and (4, 2) in the uv-plane under T.
2. Consider a transformation T defined by $x = \sqrt{v - u}$, $y = v + u$. Find the images of the points (1, 1), (1, 3), and $(\sqrt{2}, 2)$ in the xy-plane under T^{-1}.

In Problems 3–6, find the image of the set S under the given transformation.

3. S: $0 \le u \le 2$, $0 \le v \le u$; $x = 2u + v$, $y = u - 3v$
4. S: $-1 \le u \le 4$, $1 \le v \le 5$; $u = x - y$, $v = x + 2y$
5. S: $0 \le u \le 1$, $0 \le v \le 2$; $x = u^2 - v^2$, $y = uv$
6. S: $1 \le u \le 2$, $1 \le v \le 2$; $x = uv$, $y = v^2$